

Name : Shan

Sap : 72387

BSAI

OOP (Lab)

LAB Task 9

❖ **Write a Java program in static Polymorphisms.**

✚ **Q1:Simple Program**

➤ **Input**

Class EXC

```
package pkgstatic.polymorphism; // SHAN,72387,BSAI

public class StaticPolymorphism {

    public static void main(String[] args) {

        vehicle v = new vehicle();// SHAN,72387,BSAI

        v.run();

    } }
```

Class Vehicle

```
package pkgstatic.polymorphism;

public class vehicle {

    void run(){

        System.out.println("Running...."); // SHAN,72387,BSAI

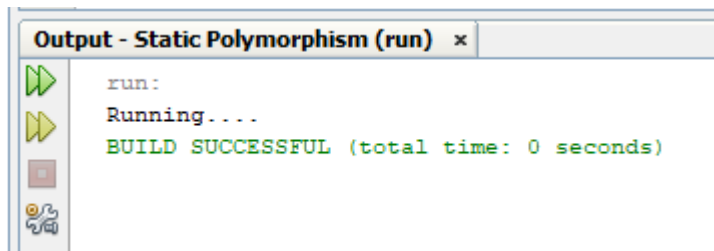
    } // SHAN,72387,BSAI

    int run(int a){

        System.out.println("Car is Running"); // SHAN,72387,BSAI

        return 0;

    } }
```



❖ Write a Java program in Dynamic Polymorphisms.

+ Q2:Simple Program

➤ Input

Class EXC

```
package run.time;

public class RunTime {

    public static void main(String[] args) {

        vehicle v= new vehicle();// SHAN,72387,BSAI

        v.run();

        Bike b = new Bike();// SHAN,72387,BSAI

        b.run();

    } }
```

Vehicle Class

```
package run.time;

class vehicle {

    void run(){

        System.out.println("Running....."); // SHAN,72387,BSAI

    } }
```

Bike Class

```

package run.time;

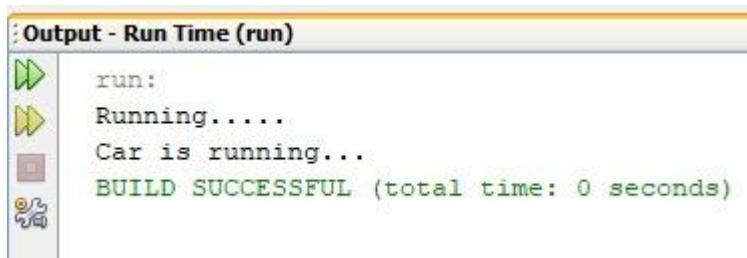
public class Bike {

    void run(){

        System.out.println("Car is running..."); // SHAN,72387,BSAI

    } }

```



- ❖ **Q2:** Write a Java program to create a base class Shape with methods draw() and calculateArea(). Create two subclasses Circle and Cylinder. Override the draw() method in each subclass to draw the respective shape. In addition, override the calculateArea() method in the Cylinder subclass to calculate and return the total surface area of the cylinder.

Input

Class Shapes

```

package shapes;

public class Shapes {

    public void draw() {

        System.out.println("Drawing a shape"); // SHAN,72387,BSAI

    }

    public double calculateArea() {

        return 0;

    } }

```

Class Circle

```

package shapes;

public class Circle extends Shapes {

    double radius; // SHAN,72387,BSAI

```

```

        public Circle(double radius) {

            this.radius = radius; // SHAN,72387,BSAI

        }

        public void draw() {

            System.out.println("Drawing a Circle");

        }

        public double calculateArea() {

            return Math.PI * radius * radius; // SHAN,72387,BSAI

        }
    }
}

```

Class Cylinder

```

package shapes;

public class Cylinder extends Shapes {

    double radius; // SHAN,72387,BSAI

    double height;

    public Cylinder(double radius, double height) {

        this.radius = radius; // SHAN,72387,BSAI

        this.height = height;

    }

    public void draw() {

        System.out.println("Drawing a Cylinder"); // SHAN,72387,BSAI

    }

    public double calculateArea() {

        return 0; // SHAN,72387,BSAI

    }
}

```

Class Main

```

package shapes;

public class ShapesTest {

    public static void main(String[] args) {

```

```
Shapes myCircle = new Circle(5.0); // SHAN,72387,BSAI

myCircle.draw();

System.out.println("Circle Area: " + myCircle.calculateArea());

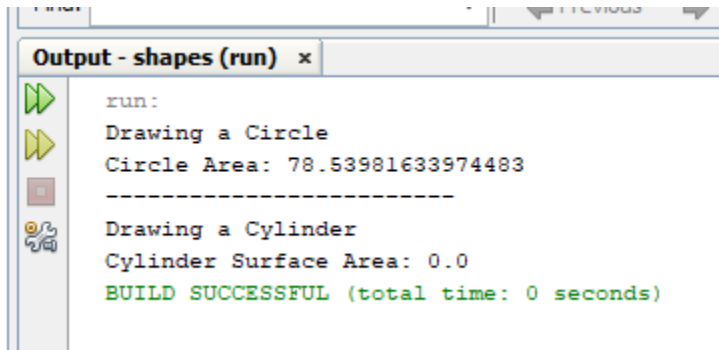
System.out.println("-----");// SHAN,72387,BSAI

Shapes myCylinder = new Cylinder(5.0, 10.0);

myCylinder.draw();// SHAN,72387,BSAI

System.out.println("Cylinder Surface Area: " + myCylinder.calculateArea());

}}
```



```
Output - shapes (run) x
run:
Drawing a Circle
Circle Area: 78.53981633974483
-----
Drawing a Cylinder
Cylinder Surface Area: 0.0
BUILD SUCCESSFUL (total time: 0 seconds)
```